

RC Cantilever Column: Lattice Modelling with Force-Based Beam–Column and 2D Continuum References

Models, calibration, and static / dynamic analyses

Units: kip, inch, second

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1 Overview

This report documents one physical RC cantilever column (Section 2) modelled four ways and analysed under static (pushover) and dynamic (seismic time-history) loading. The primary model is the **RC lattice**; three **reference** idealizations of the same specimen are used for comparison.

Table 1: Models. All share the specimen geometry, material grades, and reinforcement.

Model	Description
RC lattice (primary)	2D axial-strut assembly on a structured <code>gmsh</code> grid; reinforcement as steel struts on shared nodes (Section 3.1).
Fiber beam-column (subdivided)	Force-based <code>forceBeamColumn</code> fiber column, subdivided into 96 elements (dynamic reference; Section 3.2).
Fiber beam-column (single)	The same fiber column as a single <code>forceBeamColumn</code> element with 5 integration points (Section 3.3).
2D continuum	Plane-stress quads with <code>ASDConcrete3D + PlaneStress</code> , longitudinal bars as steel struts (Section 3.4).

Table 2: Analysis matrix. Each case compares the RC lattice against the listed reference.

Script	Analysis	Constitutive	Reference(s)	Excitation
<code>pushover_linear.py</code>	pushover	Elastic	fiber beam-column	—
<code>pushover.py</code>	pushover	Concrete02 + Steel02	fiber beam-column; 2D continuum	—
<code>dynamic_linear.py</code>	dynamic	Elastic	fiber beam-column	sine; El Centro
<code>dynamic.py</code>	dynamic	Concrete02 + Steel02	fiber beam-column; 2D continuum	sine; El Centro
<code>single_beamcolumn.py</code>	pushover	Concrete02 + Steel02	single-element fiber beam-column	—
	dynamic	Concrete02 + Steel02	single-element fiber beam-column	sine; El Centro

The lattice concrete strut area is calibrated so the lattice and the chosen reference share the same initial lateral stiffness K_0 (Section 5); this same calibration fixes the fundamental period used by the dynamic cases.

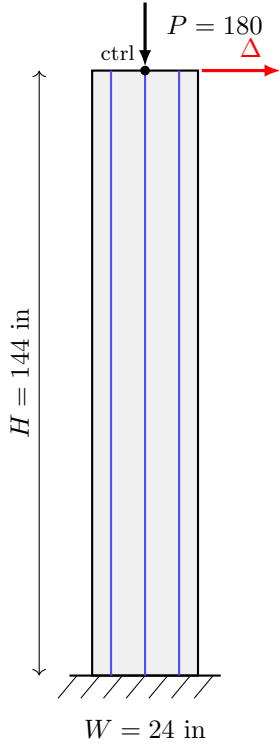
2 Physical specimen

A vertical RC cantilever, fixed at the base and free at the top, carrying a constant axial compression at the top and loaded laterally in the plane of its larger section dimension. The specimen is identical for every model and analysis case.

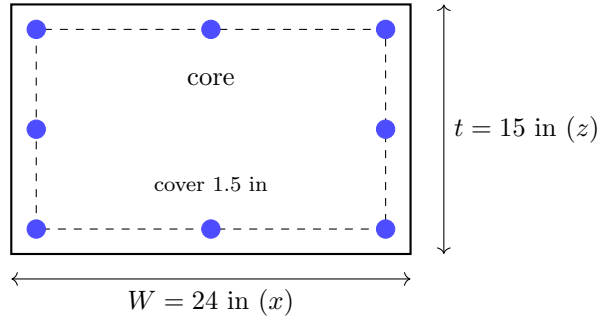
2.1 Geometry and loading

Table 3: Specimen geometry and applied loads.

Quantity	Symbol	Value
Cantilever height	H	144 in
Section depth (in-plane, x)	W	24 in
Section width (out-of-plane)	t	15 in
Clear cover	–	1.5 in
Axial load (constant, top)	P	180 kip (compression)
Reference lateral load shape	H_{ref}	10 kip (distribution only)
Base support	–	fully fixed (u_x, u_y)
Gravity constant	g	386.4 in/s ²



(a) Elevation (schematic).



(b) Cross-section: $A_s = 1.8/1.2/1.8 \text{ in}^2$ at $x = -10.5/0/+10.5$ in.

Figure 1: Physical specimen. Lateral load and displacement Δ act along x (bending about the z axis). Drawings are schematic; dimensions are labelled with their true values.

2.2 Materials

Concrete is defined by two physical grades — a *confined core* and an *unconfined cover* — and the reinforcement by a single steel grade. Values are stored as positive magnitudes; the OpenSees sign convention (compression negative) is applied in the material mapping (Section 4).

Table 4: Concrete grades (physical input). ρ is mass density.

Grade	E (ksi)	ν	ρ (kip s ² /in ⁴)	f'_c (ksi)	ε_{c0}	f_{cu} (ksi)	ε_U	f_t (ksi)
Confined core	4030	0.20	2.25×10^{-7}	6.0	0.004	4.0	0.030	0.6
Unconfined cover	4030	0.20	2.25×10^{-7}	5.0	0.002	0.0	0.006	0.5

The Concrete02 softening shape factor is $\lambda = 0.1$ for both grades.

Table 5: Reinforcing-steel grade (Steel02 / Giuffr -Menegotto-Pinto).

Grade	f_y (ksi)	E_0 (ksi)	b	R_0	c_{R1}	c_{R2}
Reinforcement	60.0	30000	0.01	18.0	0.925	0.15

2.3 Reinforcement layout

Table 6: Reinforcement. Longitudinal bars lump the benchmark 3+2+3 bars at 0.6 in² each into one area per layer; transverse ties are placed one per mesh row.

Type	Position	Area (in ²)	Extent
Longitudinal layer 1	$x = -10.5$ in	1.8	full height $0 \leq y \leq 144$
Longitudinal layer 2	$x = 0$ in	1.2	full height $0 \leq y \leq 144$
Longitudinal layer 3	$x = +10.5$ in	1.8	full height $0 \leq y \leq 144$
Transverse ties	each mesh row y_i	0.3 per tie	$x = -10.5 \rightarrow +10.5$

Total longitudinal steel area is 4.8 in², identical in every model. Transverse ties (stirrups) are present in the lattice only; the fiber and continuum references carry the longitudinal bars only.

3 Numerical models

3.1 RC lattice model (primary)

The lattice is a 2D assembly of axial struts (two translational DOFs per node) built on a structured `gmsh` grid; struts connect node pairs within a peridynamic-style horizon, and reinforcement is added as steel struts on the shared lattice nodes (perfectly bonded).

Table 7: RC lattice model definition.

Item	Specification
Model dimension	ndm=2, ndf=2
Mesh generator	gms h structured transfinite grid
Mesh size	1.5 in (16×96 cells)
Nodes	1649 (17×97 grid)
Strut element	corotTruss (corotational, geometric nonlinearity)
Strut connectivity	horizon rule, $r = \text{horizon} \times \text{mesh} = 1.5 \times 1.5 = 2.25$ in
orthogonal struts	length 1.5 in (vertical + horizontal)
diagonal struts	length $1.5\sqrt{2} = 2.1213$ in
Concrete material	per-zone length-regularized Concrete02 (4 distinct)
Rebar material	Steel02 (1 grade)
Base support	17 base-row nodes fixed (u_x, u_y)
Control node	top-centre node

Table 8: Lattice element inventory (7902 struts; 5 uniaxial materials).

Concrete struts	count	Reinforcement struts	count
vertical (orthogonal)	1632	longitudinal (3 layers)	288
horizontal (orthogonal)	1552	transverse ties (rows)	1358
diagonal	3072		
concrete total	6256	rebar total	1646

Length-regularized concrete (crack-band). Both softening branches of the strut concrete law are regularized by the strut length L , so that the dissipated energy is mesh-objective:

$$E_{ts} = \frac{f_t^2 L}{2G_f} \quad (\leq 0.5E), \quad \varepsilon_U = \varepsilon_{c0} + \frac{2G_{fc}}{(f_c + f_{cu})L} \quad (\geq \varepsilon_U^{\text{grade}}),$$

with a residual compression plateau $f_{cu} \geq 0.2f_c$. The fracture energies are $G_f = 4.0 \times 10^{-3}$ kip/in (tension) and $G_{fc} = 1.5$ kip/in (compression). This yields four concrete materials (two zones $\times$ two strut lengths):

Table 9: Lattice **Concrete02** arguments ($f_{pc}, \varepsilon_{c0}, f_{pcu}, \varepsilon_U, \lambda, f_t, E_{ts}$), compression negative.

Zone / strut length	Arguments
Cover, $L = 1.5$	(−5.0, −0.002, −1.0, −0.33533, 0.1, 0.5, 46.875)
Cover, $L = 2.1213$	(−5.0, −0.002, −1.0, −0.23770, 0.1, 0.5, 66.291)
Core, $L = 1.5$	(−6.0, −0.004, −4.0, −0.20400, 0.1, 0.6, 67.500)
Core, $L = 2.1213$	(−6.0, −0.004, −4.0, −0.14542, 0.1, 0.6, 95.459)
Rebar (Steel02)	(60.0, 30000, 0.01, 18.0, 0.925, 0.15)

Linear-material variant. For the elastic cases the concrete struts use **Elastic** at $E = 4030$ ksi and the rebar struts use **Elastic** at $E_0 = 30000$ ksi; the topology and node grid are unchanged.

Mass. Nodal mass is lumped by tributary volume ($\rho t A_{\text{trib}}/4$ per quad corner). The total lattice self-mass is $\sum m = 1.1664 \times 10^{-2} \text{ kip s}^2/\text{in}$ ($= \rho V$, $V = 24 \times 144 \times 15$). For the dynamic cases a top lumped mass $P/g = 0.4658 \text{ kip s}^2/\text{in}$ (the axial load as tributary seismic mass) is added at the top nodes (top/self ratio ≈ 40).

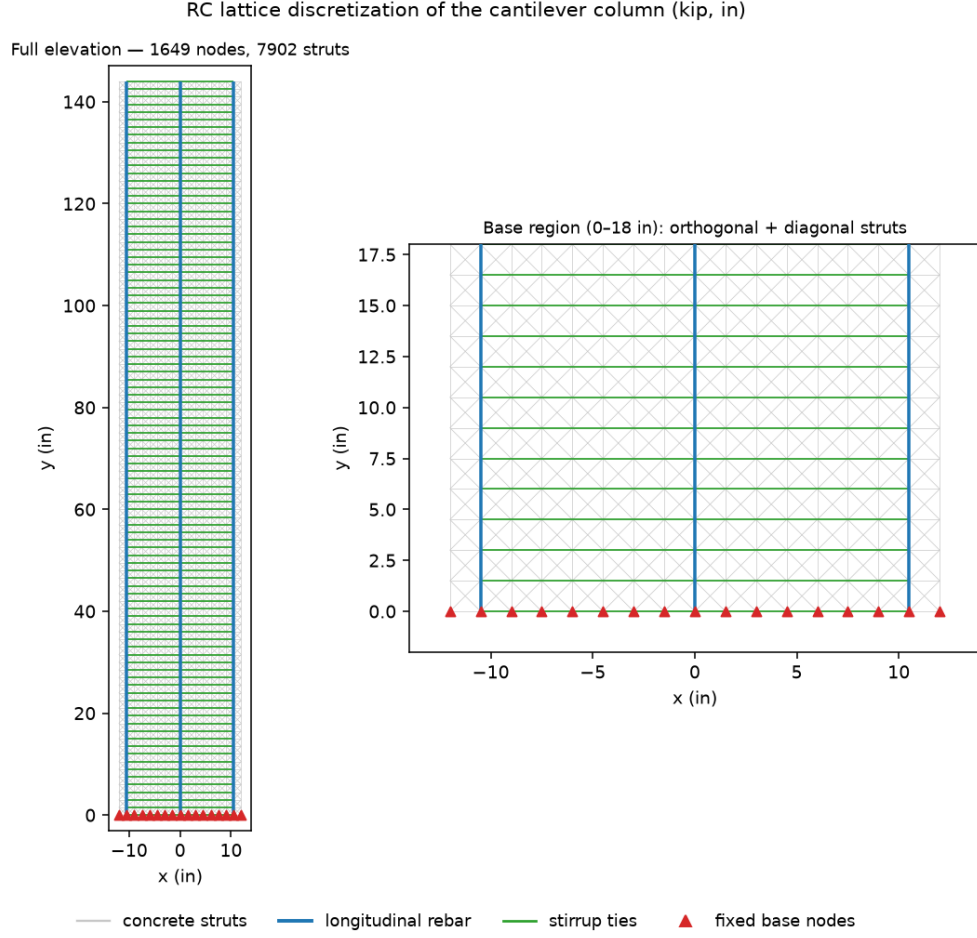


Figure 2: RC lattice discretization: concrete horizon struts (grey), longitudinal rebar struts (blue), transverse stirrup ties (green), and fixed base nodes (red). Left: full elevation; right: base region detail showing the orthogonal and diagonal strut pattern.

3.2 Force-based fiber beam–column reference (subdivided)

The dynamic reference is a force-based `forceBeamColumn` fiber column in a 2D model with three DOFs per node (u_x, u_y, θ_z), subdivided into 96 equal elements (nodes at the lattice row heights) so the self-mass distributes by tributary length identically to the lattice.

Table 10: Fiber beam–column model definition.

Item	Specification
Model dimension	<code>ndm=2, ndf=3</code>
Element	<code>forceBeamColumn</code>
Number of elements	96 (dynamic); 1 (static pushover / single-element study)
Geometric transform	<code>PDelta</code>
Beam integration	<code>Lobatto</code> , 5 integration points per element
Fiber section	24 (y) $\times$ 15 (z), cover 1.5 in
core patch	10 $\times$ 1 fibers (material: core)
cover patches	1-fiber strips (material: cover)
steel layers	3/2/3 bars at $y = -10.5/0/+10.5$, $A_s = 0.6 \text{ in}^2$
Axial load	$P = 180 \text{ kip}$ at the top node (held constant)
Self-mass (dynamic)	distributed by tributary length; top mass P/g at free top

The fiber materials are matched to the lattice grades: nonlinear cases use **Concrete02** (core/cover) and **Steel02**; linear cases use **Elastic** ($E = 4030 \text{ ksi}$ concrete, $E_0 = 30000 \text{ ksi}$ steel). The fiber-section **Concrete02** laws use the grade values directly (no length regularization; there is no fiber analog):

Table 11: Fiber-section **Concrete02** arguments ($f_{pc}, \varepsilon_{c0}, f_{pcu}, \varepsilon_U, \lambda, f_t, E_{ts}$), compression negative.

Fiber material	Arguments
Core (nonlinear)	(−6.0, −0.004, −4.0, −0.030, 0.1, 0.6, 403.0)
Cover (nonlinear)	(−5.0, −0.002, −0.0, −0.006, 0.1, 0.5, 403.0)
Steel (Steel02)	(60.0, 30000, 0.01, 18.0, 0.925, 0.15)

3.3 Single-element fiber beam–column

The single-element model (script `single_beamcolumn.py`) idealizes the fiber column as one `forceBeamColumn` element with 5 Gauss–Lobatto integration points (fixed base node, free top node), with the same fiber section and materials as Section 3.2. It serves both as the pushover reference and the dynamic reference for that script. Because it has no interior nodes, the column self-mass is carried as distributed element mass ($\text{massDens} = m_{\text{self}}/H$) and the axial-load tributary seismic mass P/g is lumped at the free top node. It is the same model OpenSees builds for the static pushover reference of `pushover.py`.

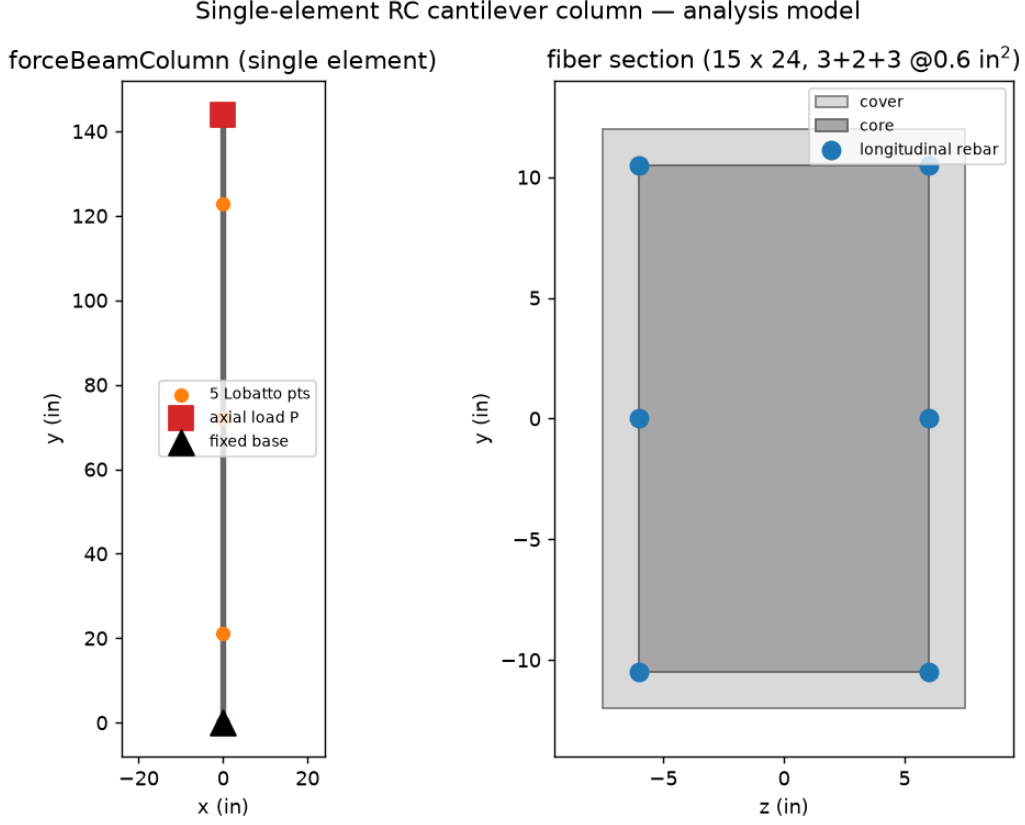


Figure 3: Single-element fiber beam-column: column elevation (5 Lobatto points, fixed base, top node) and the fiber section (core/cover patches and longitudinal rebar).

3.4 2D plane-stress continuum reference

The continuum reference is a structured plane-stress quad mesh of the same column, with nonlinear nD concrete and the longitudinal bars represented as steel struts sharing the quad nodes.

Table 12: 2D continuum model definition.

Item	Specification
Model dimension	<code>ndm=2</code> , <code>ndf=2</code> (plane stress, thickness $t = 15$ in)
Mesh generator	<code>gmsh</code> structured grid, mesh size 1.5 in
Nodes	1649
Elements	1824 total: 1536 plane-stress quads + 288 longitudinal rebar struts
Concrete material	<code>ASDConcrete3D</code> + <code>PlaneStress</code> wrapper, per zone (4 nD materials)
Rebar material	<code>Steel102</code> (1 uniaxial), longitudinal bars only (no stirrups)
Regularization	crack-band by element characteristic length $l_{ch} = 1.5$ in
Hysteresis setting	pure damage (<code>plastic_frac=0</code> , $d_{\max} = 0.95$)
Base support	17 base-row nodes fixed (u_x, u_y)
Control node	top-centre node (id 873)

`ASDConcrete3D` is driven by uniaxial tension ($Te/Ts/Td$) and compression ($Ce/Cs/Cd$) curves

given as positive magnitudes; damage values d follow from $d = \text{clip}(1 - \sigma/(E\varepsilon), 0, d_{\max})$ (pure damage, `plastic_frac` = 0):

Table 13: Continuum ASDConcrete3D curves ($E = 4030$ ksi, $\nu = 0.2$, $l_{ch} = 1.5$ in). Strain / stress (ksi) / damage triples.

Branch	Core	Cover
Te (strain)	0, 1.49×10^{-4} , 9.038×10^{-3}	0, 1.24×10^{-4} , 1.0791×10^{-2}
Ts (ksi)	0, 0.6, 0.012	0, 0.5, 0.010
Td	0, 0, 0.95	0, 0, 0.95
Ce (strain)	0, 5.96×10^{-4} , 4.0×10^{-3} , 0.204	0, 4.96×10^{-4} , 2.0×10^{-3} , 0.402
Cs (ksi)	0, 2.4, 6.0, 4.0	0, 2.0, 5.0, 0.0
Cd	0, 0, 0.6278, 0.95	0, 0, 0.3797, 0.95

4 Material mapping summary

Table 14: Physical grade $\rightarrow$ per-model OpenSees material.

Use	Nonlinear	Linear (elastic)
Lattice concrete strut	length-regularized Concrete02	Elastic (E)
Lattice / fiber / quad rebar	Steel02	Elastic (E_0)
Fiber-section concrete	plain Concrete02 (grade values)	Elastic (E)
Continuum concrete	ASDConcrete3D + PlaneStress	—

5 Stiffness calibration

The single calibration parameter is the concrete strut cross-sectional area A (uniform over all concrete struts), chosen so the lattice’s initial lateral stiffness K_0 equals the chosen reference’s. The reinforcement areas are fixed by the specimen and are not calibrated.

- **Nonlinear cases:** a plain elastic lattice (reinforcement omitted) has $K_0 \propto A$, so a single solve gives A from the reference K_0 .
- **Linear cases:** the full RC elastic lattice (concrete + rebar) has a stiffness that is monotone but not proportional in A (rebar adds a parallel stiffness), so A is found by a secant root-find on $K_0(A) - K_0^{\text{ref}}$.

Table 15: Calibration results. K_0^{ref} is the reference’s initial lateral stiffness (first finite pushover step).

Configuration	Reference	K_0^{ref} (kip/in)	Calibrated A (in ²)
Linear	fiber beam-column	68.69	11.736
Nonlinear	fiber BC (single & subdivided)	61.53	12.453
Nonlinear	2D continuum	80.34	16.260

For the linear calibration the resulting lattice $K_0 = 68.69$ kip/in (-0.00% vs. the reference). The continuum reference yields a larger A (and a stiffer, shorter-period lattice) than the fiber reference because K_0^{ref} is larger.

6 Analysis types and parameters

6.1 Pushover

A constant-gravity stage (axial P) is applied first and held; the lateral displacement of the control node is then increased monotonically under displacement control while the base shear (sum of horizontal base reactions) is recorded.

Table 16: Pushover analysis parameters (all models).

Item	Setting
Constraint handler	Transformation
DOF numberer	RCM
System of equations	BandGeneral
Gravity stage	LoadControl , 10 steps; Newton ; NormDispIncr 10^{-8}
Hold gravity	loadConst (time reset to 0)
Lateral stage	DisplacementControl on control DOF, $dU = 0.05$ in
Target tip displacement	10.0 in
Convergence test	NormDispIncr , tol 10^{-5} (lattice), 10^{-6} (fiber)
Primary algorithm	Newton
Recovery on failure	sub-steps 5/20/100 with Newton → KrylovNewton
Base shear	$-\sum$ horizontal base reactions

6.2 Dynamic (seismic time history)

After the held-gravity stage, the base is driven by a **UniformExcitation** (ground acceleration record) along x . The record intensity is tuned on the (inexpensive) fiber column to a target peak roof drift; the *same* scaled record then drives the lattice (and, for the continuum cases, the continuum). Two records are used per case: a harmonic acceleration tuned to resonance with the lattice fundamental period T_1 , and the recorded El Centro 1940 NS ground motion.

Table 17: Dynamic analysis parameters (all models).

Item	Setting
Constraint handler / numberer / system	Transformation / RCM / BandGeneral
Mass	lumped self-mass + top mass P/g
Damping	modal damping , 5% at modes 1 and 2
Time integrator	HHT , $\alpha = 0.7$ (numerical damping)
Integration step	$\Delta t = 0.005$ s (nonlinear); 0.01 s (linear)
Convergence test	NormDispIncr , tol 10^{-6} , 50 iterations
Primary algorithm	Newton
Recovery on failure	sub-steps 10/50/200 with Newton → KrylovNewton → NewtonLineSearch(Bisection) → ModifiedNewton
Excitation pattern	UniformExcitation along x
Target peak drift (tuning)	1.0 in (linear); 2.0 in (nonlinear)

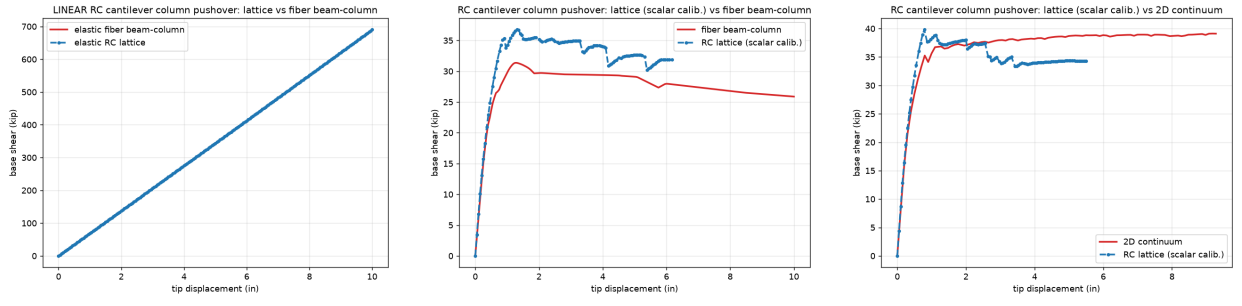
Table 18: Base excitation records. The sine period equals the lattice T_1 (resonant); the continuum sine case is capped at 8 cycles.

Record	Points	Δt_{rec} (s)	Peak (g)	Duration (s)
Sine resonance (20 cycles, 100 samples/cycle)	2001	$T_1/100$	1.000	$20 T_1$
Sine resonance (8 cycles, continuum)	801	$T_1/100$	1.000	$8 T_1$
El Centro 1940 NS	1559	0.0200	0.319	31.2

The continuum dynamic reference uses **ASDConcrete3D** configured as pure damage (`plastic_frac = 0`); it is computationally heavy (≈ 1.5 s/step over 1536 quads), hence the 8-cycle sine cap.

7 Results

7.1 Pushover



(a) Linear (elastic), fiber reference.

(b) Nonlinear, fiber reference.

(c) Nonlinear, continuum reference.

Figure 4: Base shear versus tip displacement: RC lattice (blue, dashed) and the reference (red).

Table 19: Pushover results. Peak shear is the maximum recorded base shear; max tip displacement is the last converged step.

Configuration	Model	Peak shear (kip)	Max tip disp. (in)	Converged
Linear (fiber ref.)	Fiber beam-column	686.9 (@10 in)	10.00	yes
	RC lattice	690.2 (@10 in)	10.00	yes
Nonlinear (fiber ref.)	Fiber beam-column	31.37	10.00	yes
	RC lattice	36.75	6.17	no (terminated)
Nonlinear (continuum ref.)	2D continuum	39.12	9.28	no (terminated)
	RC lattice	39.80	5.50	no (terminated)

7.2 Dynamic — linear (elastic), fiber reference

Sine resonance (scale $0.039\times$, $\Delta t = 0.01$ s).

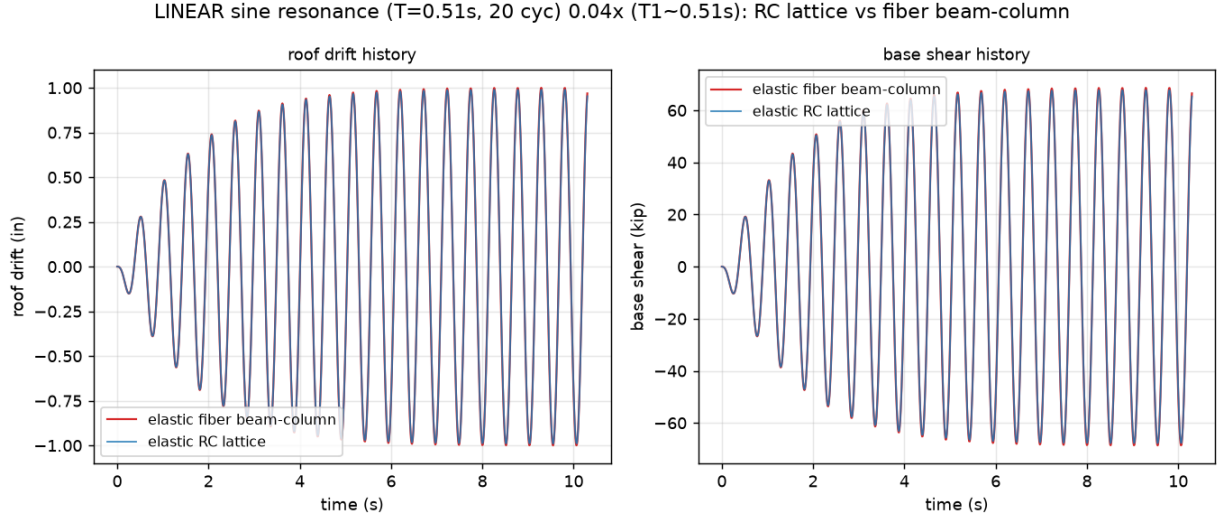


Figure 5: Linear, sine — response histories (roof drift, base shear): RC lattice vs. elastic fiber beam-column.

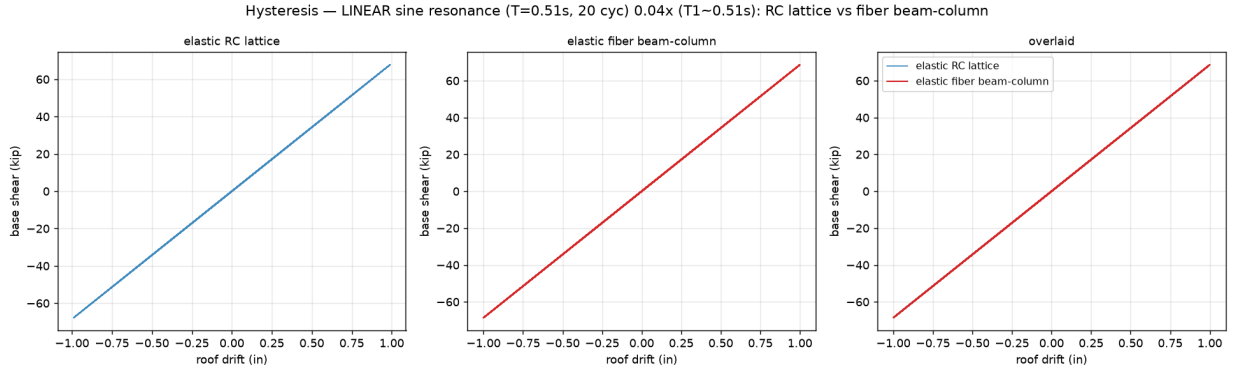


Figure 6: Linear, sine — hysteresis: RC lattice (left), fiber beam-column (middle), overlaid (right).

El Centro 1940 NS (scale 0.413 $\times$, $\Delta t = 0.01$ s).

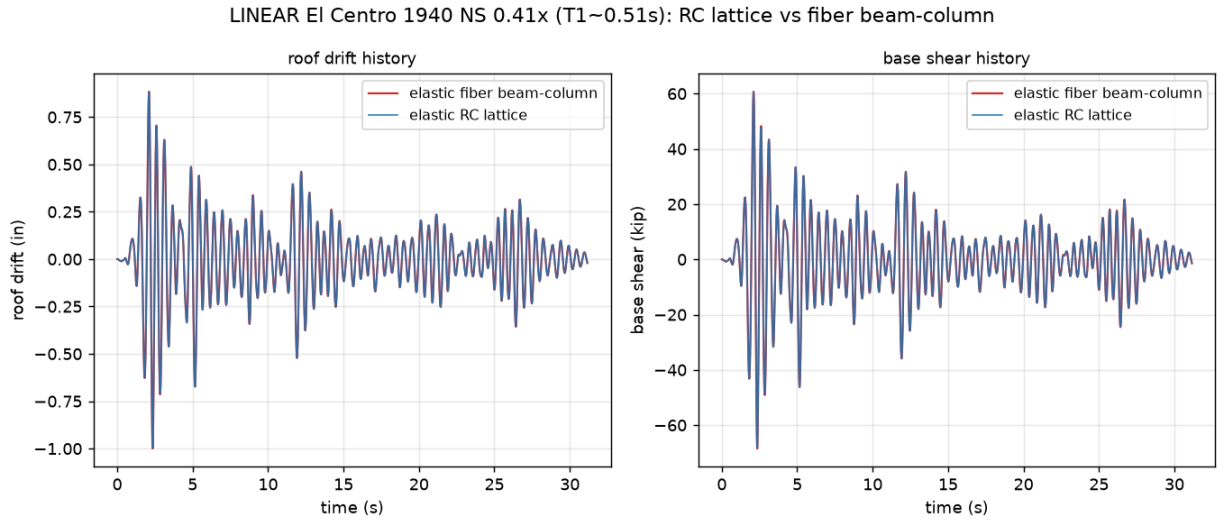


Figure 7: Linear, El Centro — response histories (roof drift, base shear).

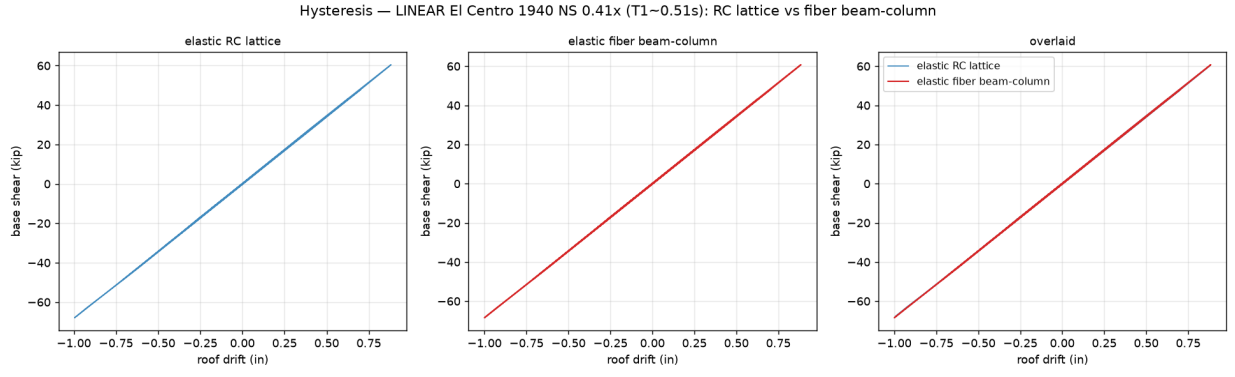


Figure 8: Linear, El Centro — hysteresis: RC lattice (left), fiber beam-column (middle), overlaid (right).

7.3 Dynamic — nonlinear, fiber beam-column reference

Sine resonance (scale $0.154\times$, $\Delta t = 0.005$ s).

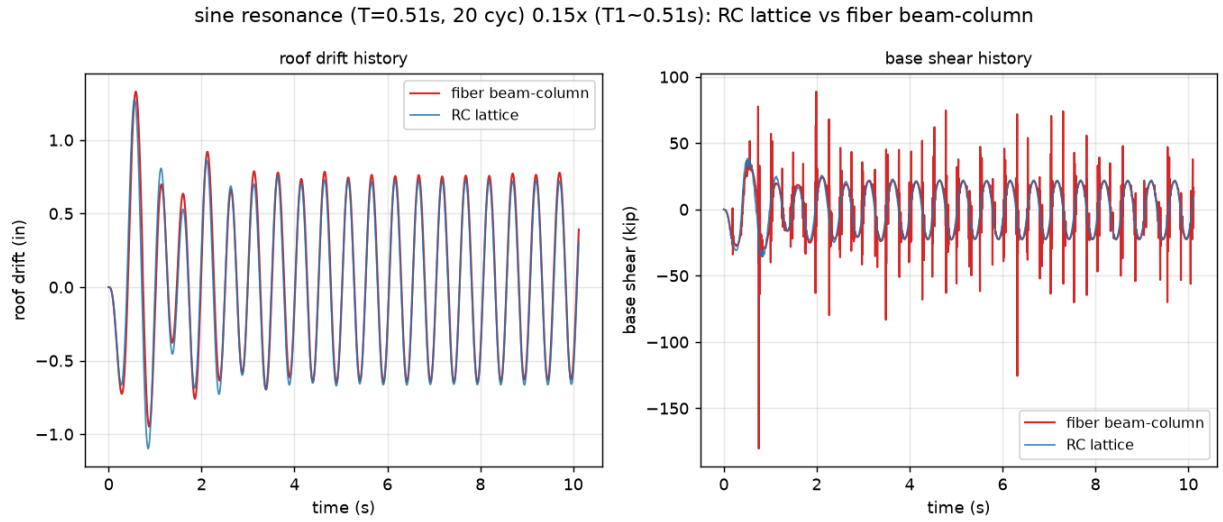


Figure 9: Nonlinear, sine — response histories: RC lattice vs. fiber beam-column.

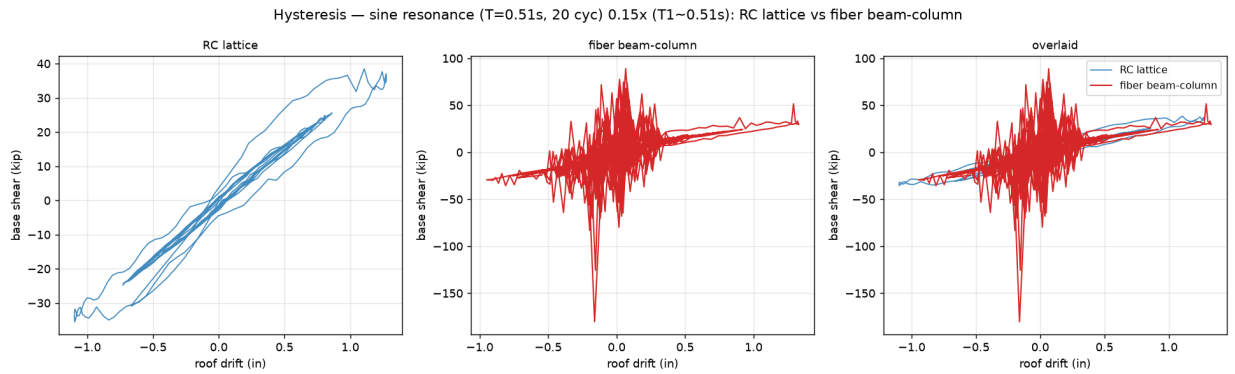


Figure 10: Nonlinear, sine — hysteresis: RC lattice (left), fiber beam-column (middle), overlaid (right). Note the separate vertical scales.

El Centro 1940 NS (scale $0.545\times$, $\Delta t = 0.005$ s).

El Centro 1940 NS 0.54x (T1~0.51s): RC lattice vs fiber beam-column

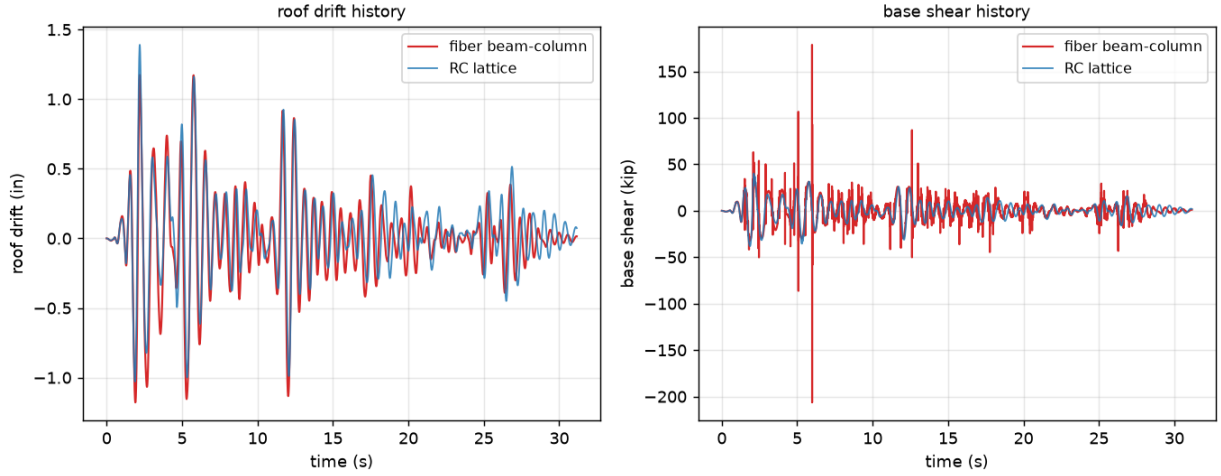


Figure 11: Nonlinear, El Centro — response histories.

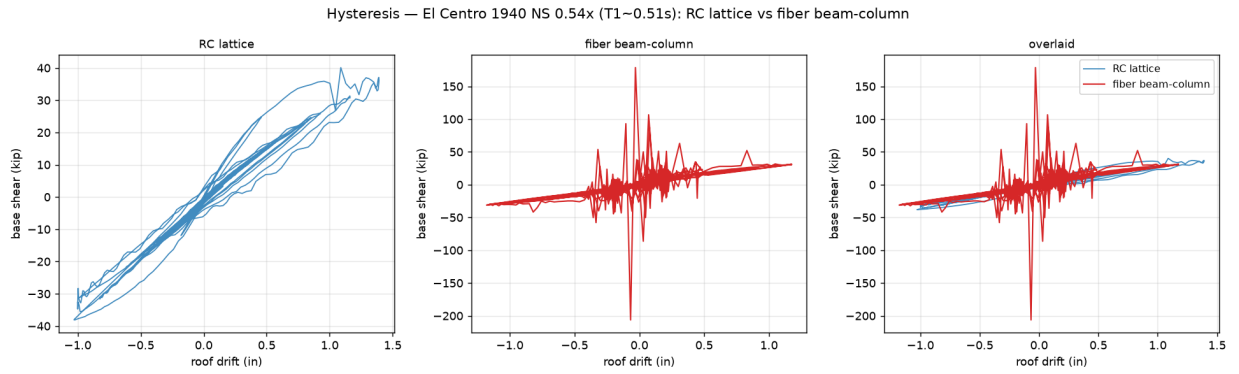


Figure 12: Nonlinear, El Centro — hysteresis: RC lattice (left), fiber beam-column (middle), overlaid (right). Note the separate vertical scales.

7.4 Dynamic — nonlinear, 2D continuum reference

Sine resonance (scale 0.312 $\times$, 8 cycles, $\Delta t = 0.005$ s).

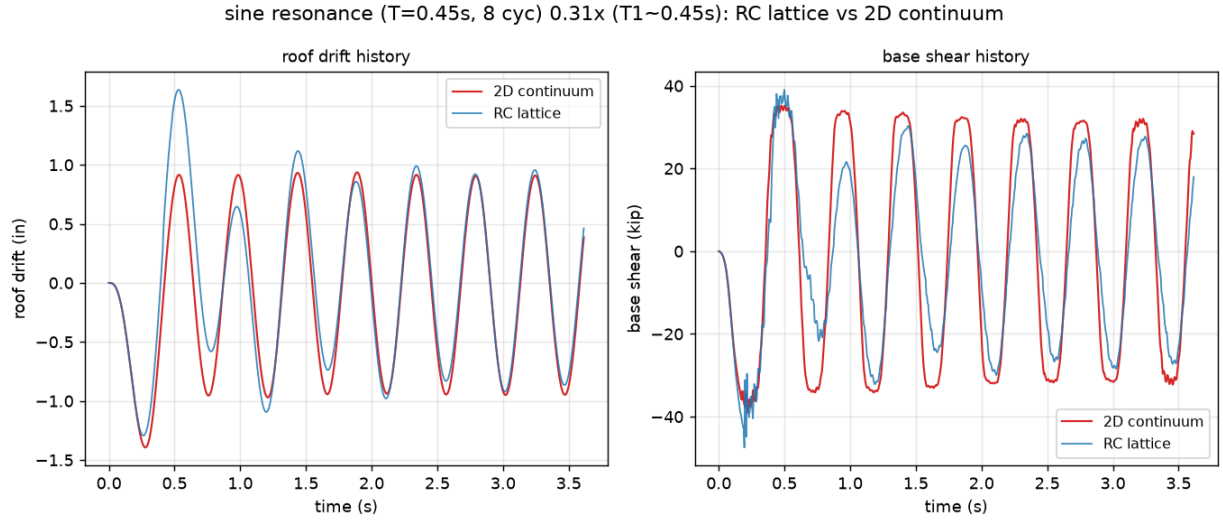


Figure 13: Nonlinear, sine — response histories: RC lattice vs. 2D continuum.

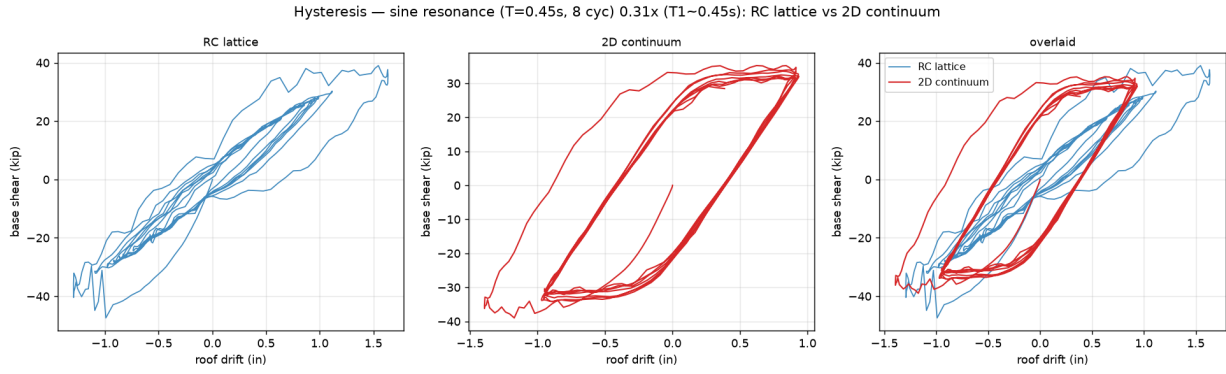


Figure 14: Nonlinear, sine — hysteresis: RC lattice (left), 2D continuum (middle), overlaid (right).

El Centro 1940 NS (scale $0.545x$, $\Delta t = 0.005$ s).

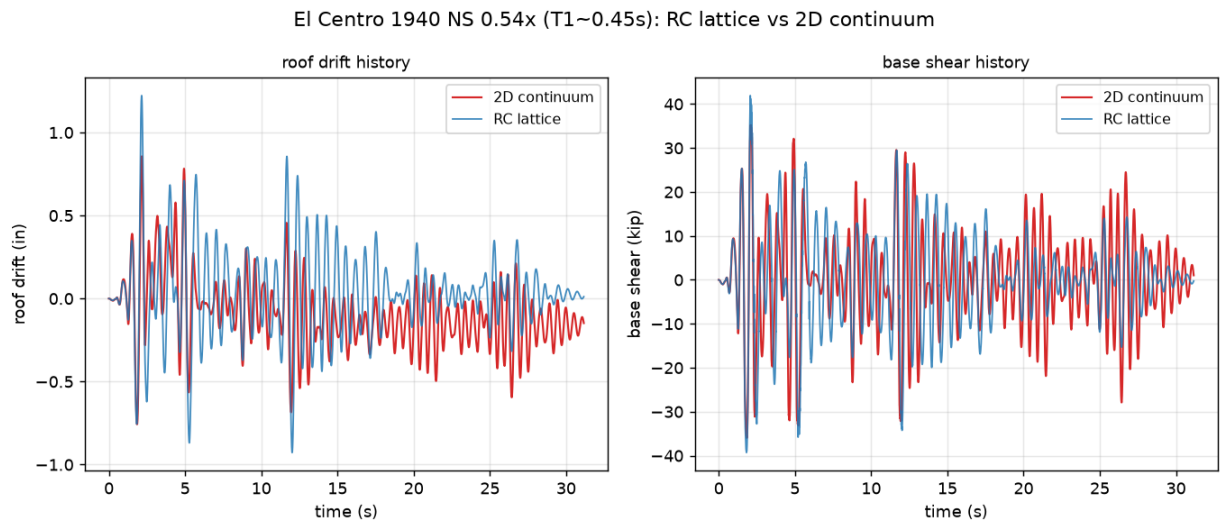


Figure 15: Nonlinear, El Centro — response histories.

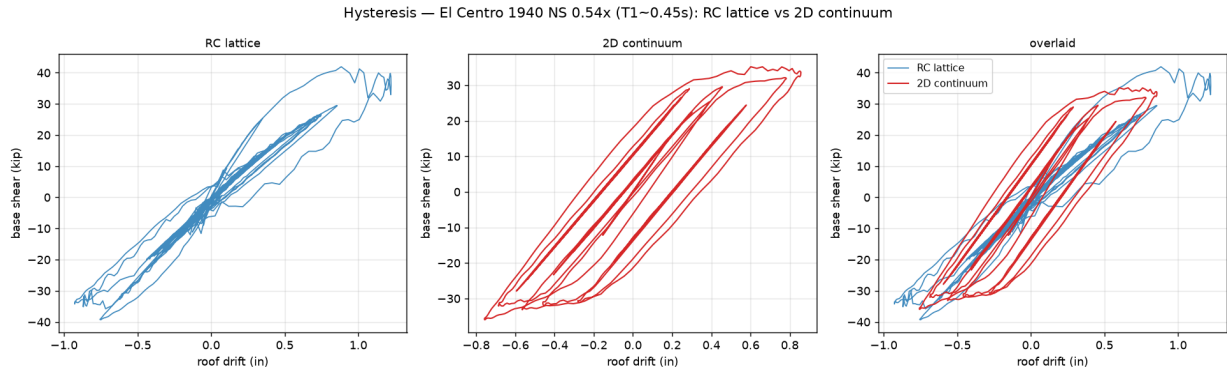


Figure 16: Nonlinear, El Centro — hysteresis: RC lattice (left), 2D continuum (middle), overlaid (right).

7.5 Single-element beam-column study

Script `single_beamcolumn.py` compares the RC lattice against the single-element fiber beam-column (Section 3.3) under both a nonlinear pushover (same machinery as Section 6.1) and a nonlinear seismic time-history (Section 6.2, $\Delta t = 0.005$ s).

Pushover.

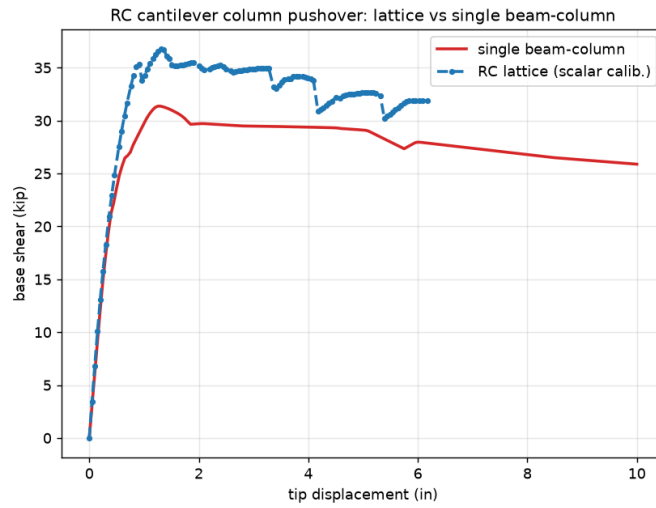


Figure 17: Pushover — base shear versus tip displacement: RC lattice vs. single-element fiber beam-column.

Sine resonance (scale $0.312\times$).

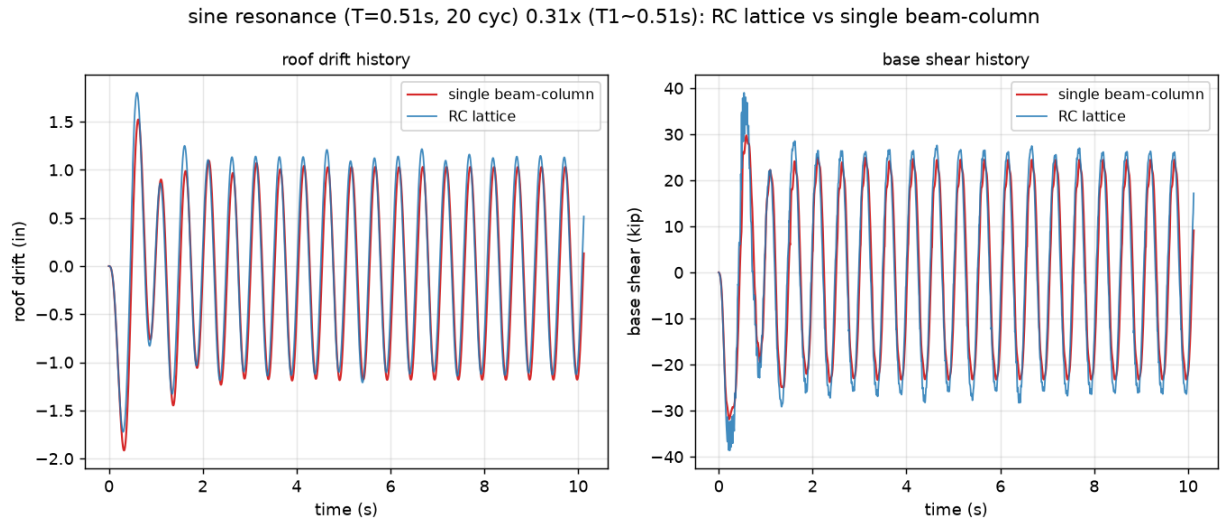


Figure 18: Single-element, sine — response histories.

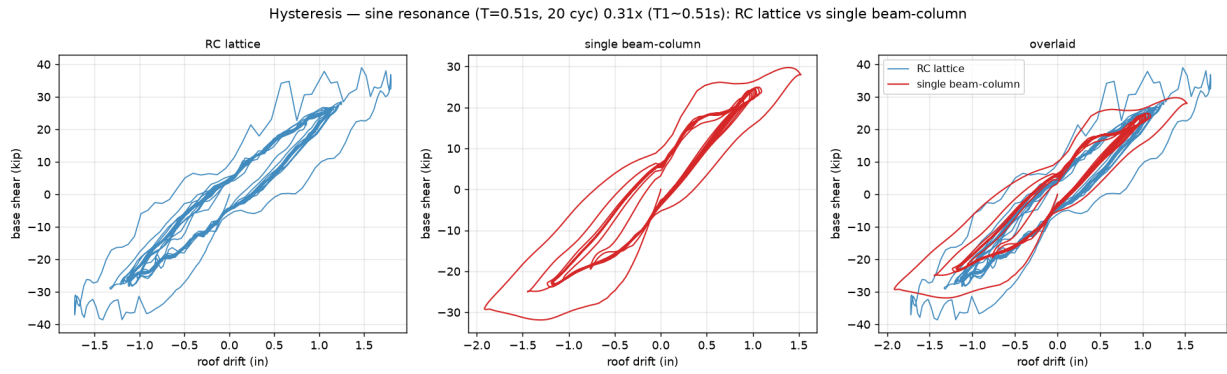


Figure 19: Single-element, sine — hysteresis: RC lattice (left), single beam-column (middle), overlaid (right).

El Centro 1940 NS (scale $0.750\times$).

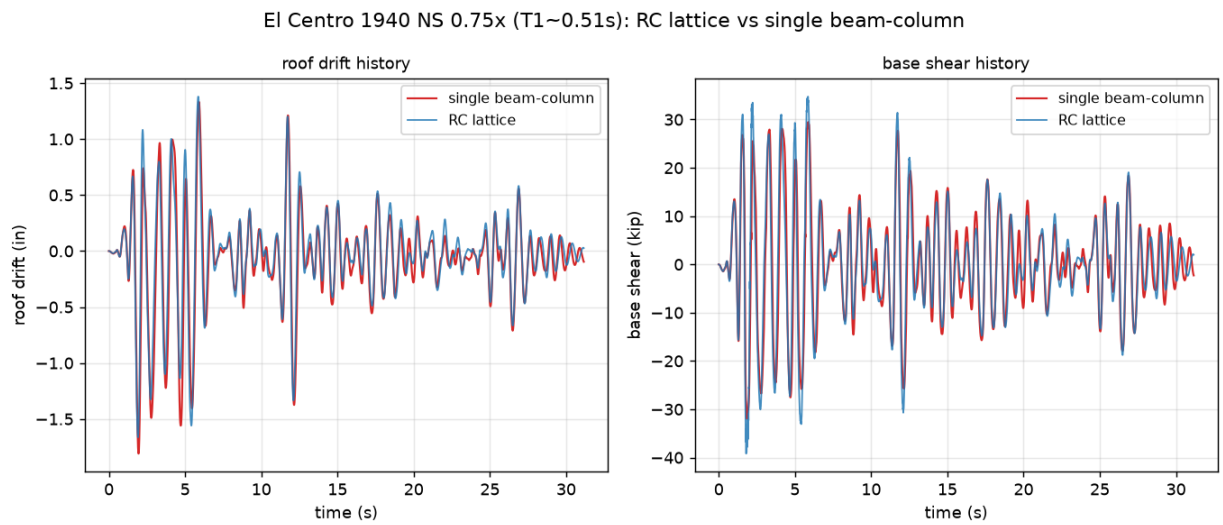


Figure 20: Single-element, El Centro — response histories.

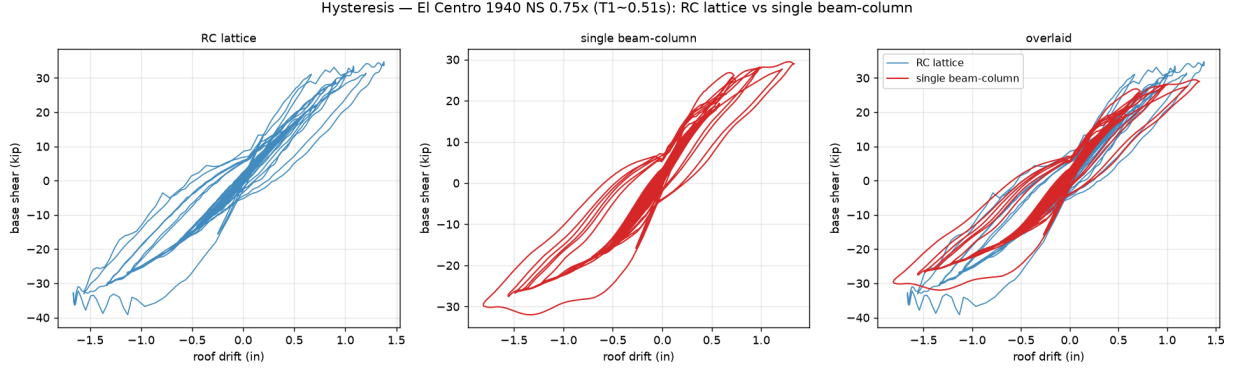


Figure 21: Single-element, El Centro — hysteresis: RC lattice (left), single beam-column (middle), overlaid (right).

7.6 Tabulated dynamic results

Table 20: Dynamic results. T_1 is the lattice fundamental period (and the sine excitation period); drift is the relative roof displacement; peak $|V|$ is the maximum absolute recorded base shear (includes transient reaction peaks); residual is the end-of-record roof drift.

Study / record	Scale	T_1 (s)	Peak drift (in)		Peak $ V $ (kip)		Residual (in)	
			Ref.	Lattice	Ref.	Lattice	Ref.	Lattice
Linear, sine	0.039×	0.515	1.000	0.990	68.7	67.8	+0.968	+0.953
Linear, El Centro	0.413×	0.515	1.000	0.995	68.5	68.0	−0.021	−0.020
Nonlinear (fiber), sine	0.154×	0.506	1.329	1.274	180.4	38.5	+0.392	+0.304
Nonlinear (fiber), El Centro	0.545×	0.506	1.179	1.391	206.3	40.1	+0.015	+0.073
Nonlinear (continuum), sine	0.312×	0.452	1.394	1.636	39.0	47.4	+0.386	+0.462
Nonlinear (continuum), El Centro	0.545×	0.452	0.857	1.222	36.0	41.9	−0.148	+0.011
Single BC, sine	0.312×	0.506	1.915	1.801	31.9	39.0	+0.132	+0.516
Single BC, El Centro	0.750×	0.506	1.809	1.664	32.0	39.2	−0.096	+0.027

The fiber-proxy fundamental period used to tune the intensity is $T_1^{\text{fiber}} = 0.520$ s (linear), 0.558 s (nonlinear subdivided), and 0.559 s (single element).

A Software and reproduction

Table 21: Software environment (native arm64, macOS).

Component	Version
Python	3.12.13 (arm64)
openseespy	3.8.0.0 (openseespymac 3.8.0.0)
gmsh	4.15.2
numpy / scipy	2.4.6 / 1.17.1
matplotlib	3.11.0

Run from the `src/` directory with the project virtual environment active:

```

python examples/column/pushover_linear.py
python examples/column/pushover.py          --reference beamcolumn
python examples/column/pushover.py          --reference continuum
python examples/column/dynamic_linear.py    --excitation sine
python examples/column/dynamic_linear.py    --excitation elcentro
python examples/column/dynamic.py           --reference beamcolumn --excitation sine
python examples/column/dynamic.py           --reference beamcolumn --excitation elcentro
python examples/column/dynamic.py           --reference continuum --excitation sine
python examples/column/dynamic.py           --reference continuum --excitation elcentro
python examples/column/single_beamcolumn.py --analysis pushover
python examples/column/single_beamcolumn.py --analysis dynamic --excitation sine
python examples/column/single_beamcolumn.py --analysis dynamic --excitation elcentro
python examples/doc/make_model_figures.py   # lattice mesh figure

```

Drawings of every analysis model are written next to each result (the *.model.png files); add --draw where the script supports it.